Influence of Visual, Proprioceptive, and Footwear Alterations on Center-of-Mass Stabilization During Perturbed Standing Balance

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Introduction

The ability to rapidly stabilize the center of mass (CoM) after an unexpected perturbation is essential for maintaining balance and preventing falls [1,2]. CoM stabilization time is influenced by sensory information, and footwear properties, yet their specific contribution remains unclear. Understanding how sensory conditions and footwear stiffness affect stabilization dynamics in healthy adults is necessary to develop CoM-based metrics applicable to clinical populations.

Methods

Seven healthy adults participated in an experiment involving unexpected anterior–posterior (AP) mechanical perturbations at the pelvis. Perturbations were applied through a custom-built motorized device using inextensible cables connected near the CoM, and participants were instructed to maintain standing balance with their arms crossed. An individualized calibration determined the perturbation intensity that elicited a compensatory step; for the experiment, we set the perturbation level to 50% of this value. Each participant completed 18 trials, combining 3 visual conditions (eyes open-EO, eyes closed-EC, incorrect visual information via virtual reality headset-VR), 2 proprioceptive conditions (firm ground or foam), and 3 footwear conditions (shoes only, ankle-foot orthoses-AFO with low stiffness, and AFO with high stiffness). Each trial included 12 perturbations. Whole-body kinematics (excluding arms) and ground reaction forces were recorded at 100 Hz using a Vicon motion capture system. CoM position and velocity were computed using subject-specific personalized biomechanical models based on OpenSim Gait2392. Stabilization time (ST) was defined as the first instant at which CoM velocity remained within three times the standard deviation of its steady-state value for at least 500 ms.

Results and discussion

Contrary to our expectations, EC and VR generally reduced ST relative to EO (Figure 1). These decreases depended on the type of surface, varying between 1–11% with firm ground and 4–17% on foam. A reason for that might be that limited or unreliable vision may have led participants to increase baseline muscle contraction as an anticipatory strategy, shortening stabilization times. Foam surface generally increased ST compared to firm ground in EO trials, with increases around 1–10%. However, under EC and VR trials, foam produced more variable effects, suggesting that when visual information is compromised, the reliance on somatosensory cues becomes more sensitive to footwear. Footwear also modulated stabilization performance. The LS AFO produced ST increases (3,2–18%) with respect to shoes only condition, while the HS AFO showed the larger deviations (11–32%) from shoes only, consistent with the idea that unfamiliar footwear makes maintaining balance more challenging.

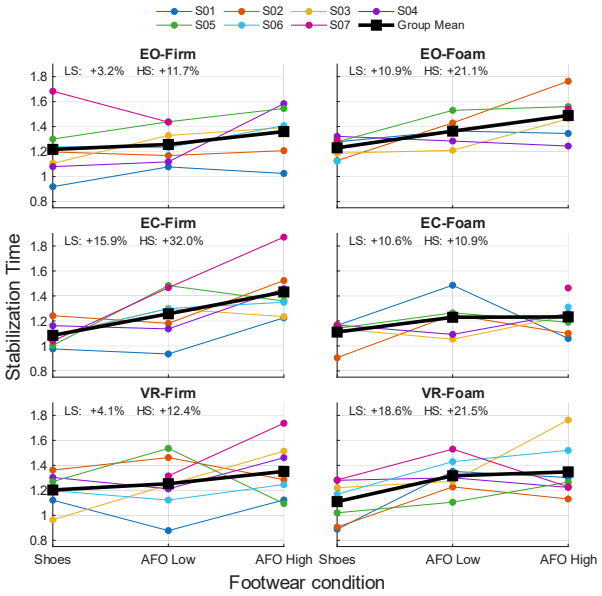


Figure 1: Mean ST for perturbations across footwear conditions and sensory combinations. Plots show group means (black) and individual subject data. Percentages indicate relative changes in ST for LS and HS AFOs compared to shoes only.

Conclusions

These preliminary results show that sensory conditions and footwear modulate CoM stabilization time in complex ways and non-intuitive patterns across vision and proprioceptive conditions. Foam generally prolonged ST when visual information was reliable, but its effects became footwear-dependent under perturbed-vision conditions. Footwear deviations from participants' habitual shoes also produced progressive changes in ST, with the HS AFO showing the largest alterations. As ST represents only one aspect of postural control, interpretation requires complementary EMG and joint-moment analyses, which will be expanded in future work.

References

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